

6. CONVOLUTION AND SAMPLING TECHNIQUES

Experiment 6.1% Convolution of Continuous-Time Signals

```
clc;
```

```
clear;
```

```
close all;
```

```
% Time axis
```

```
t = 0:0.01:5;
```

```
% Input signals
```

```
x1 = exp(-t); % Exponential signal
```

```
x2 = ones(size(t)); % Unit step signal
```

```
% Convolution
```

```
y = conv(x1,x2)*0.01;
```

```
% Time axis for output
```

```
t_conv = 0:0.01:(length(y)-1)*0.01;
```

```
% Plotting
```

```
figure;
```

```
subplot(3,1,1);
```

```
plot(t,x1,'LineWidth',2);
```

```
grid on;
```

```
title('Signal x_1(t)=e^{-t}');
```

```
xlabel('Time (s)');
```

```
ylabel('Amplitude');
```

```
subplot(3,1,2);
```

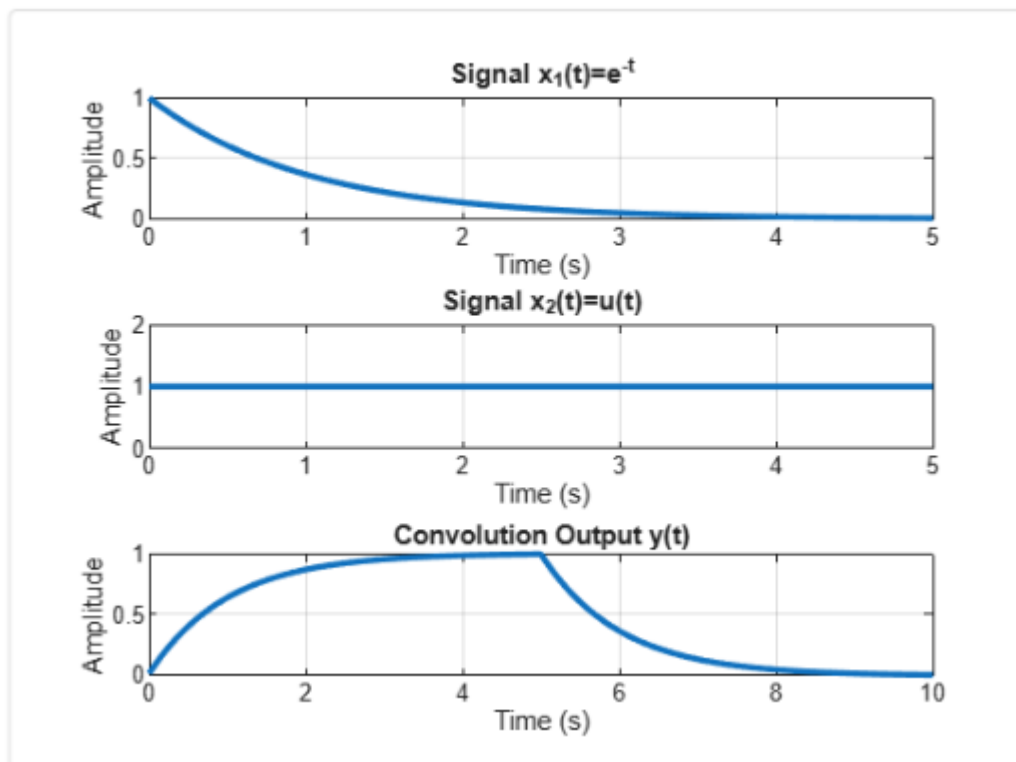
```
plot(t,x2,'LineWidth',2);
```

```
grid on;
```

```

title('Signal x_2(t)=u(t)');
xlabel('Time (s)');
ylabel('Amplitude');
subplot(3,1,3);
plot(t_conv,y,'LineWidth',2);
grid on;
title('Convolution Output y(t)');
xlabel('Time (s)');
ylabel('Amplitude');

```



Experiment 6.2

% Convolution of Discrete-Time Signals

```
clc;
```

```
clear;
```

```
close all;
```

```
% Input sequences
```

```
x1 = [1 2 3 4];
x2 = [1 1 1];
% Convolution
y = conv(x1,x2);
% Sample indices
n1 = 0:length(x1)-1;
n2 = 0:length(x2)-1;
ny = 0:length(y)-1;
% Display output
disp('Convolution Output:');
disp(y);
% Plotting
figure;
subplot(3,1,1);
stem(n1,x1,'filled');
grid on;
title('Input Sequence x_1(n)');
xlabel('n');
ylabel('Amplitude');
subplot(3,1,2);
stem(n2,x2,'filled');
grid on;
title('Input Sequence x_2(n)');
xlabel('n');
ylabel('Amplitude');
subplot(3,1,3);
```

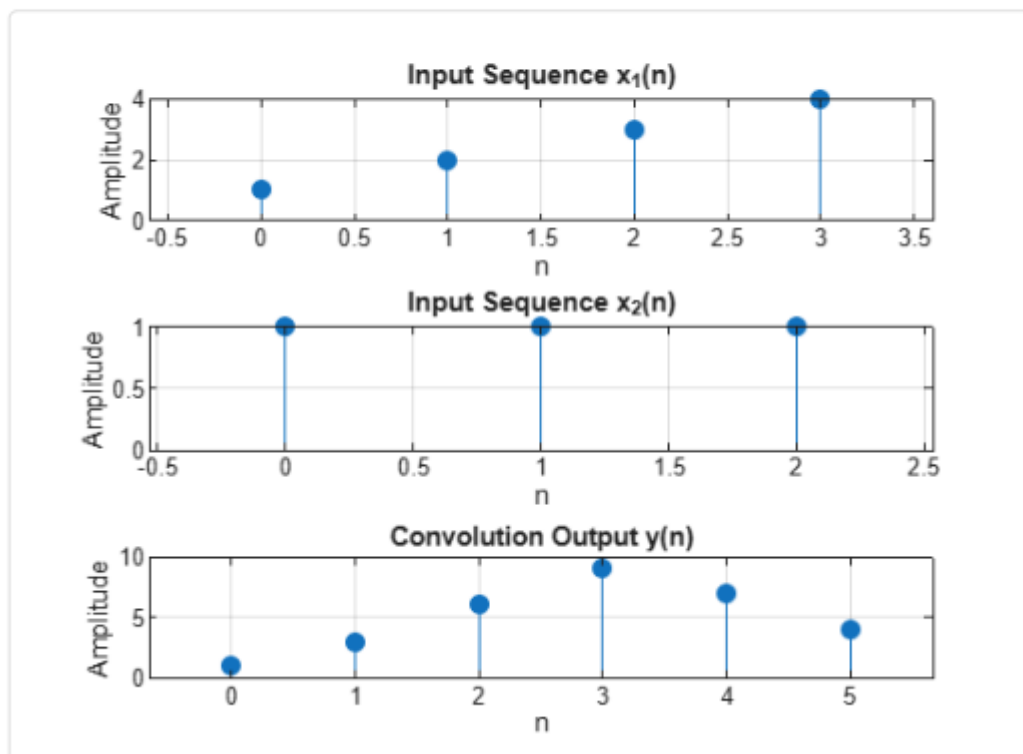
```

stem(ny,y,'filled');
grid on;
title('Convolution Output y(n)');
xlabel('n');
ylabel('Amplitude');

```

Convolution Output:

1 3 6 9 7 4



Experiment 6.3% Sampling Theorem Verification

```

clc;
clear;
close all;

% Signal parameters
fm = 5; % Signal frequency (Hz)
fs = 20; % Sampling frequency (Hz)

```

```
t = 0:0.001:1; % Continuous time
```

```
% Continuous signal
```

```
x = sin(2*pi*fm*t);
```

```
% Sampled signal
```

```
n = 0:1/fs:1;
```

```
xs = sin(2*pi*fm*n);
```

```
% Plotting
```

```
figure;
```

```
subplot(2,1,1);
```

```
plot(t,x,'LineWidth',2);
```

```
grid on;
```

```
title('Continuous-Time Signal');
```

```
xlabel('Time (s)');
```

```
ylabel('Amplitude');
```

```
subplot(2,1,2);
```

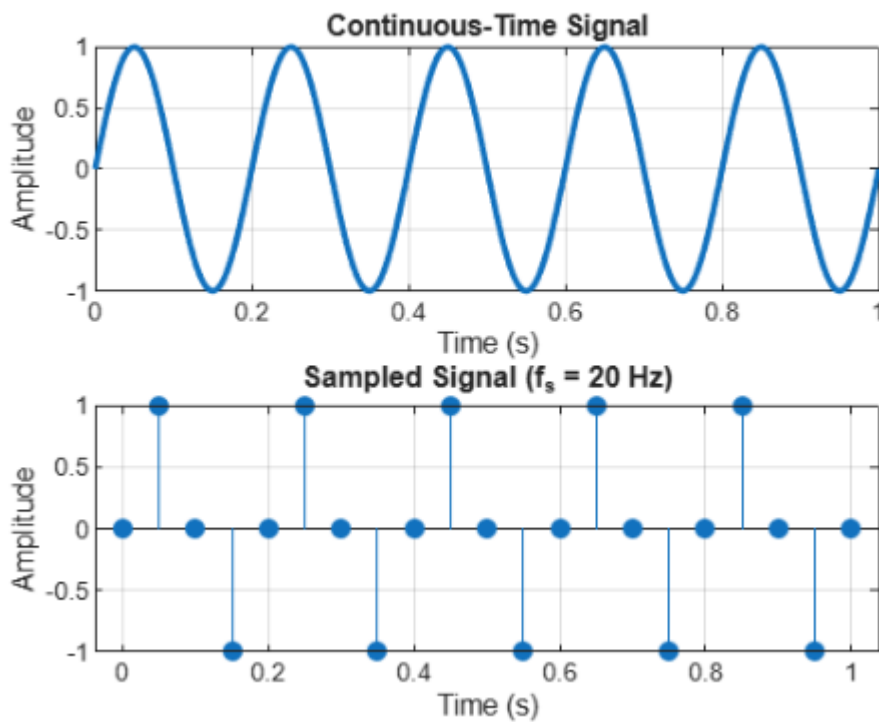
```
stem(n,xs,'filled');
```

```
grid on;
```

```
title('Sampled Signal (f_s = 20 Hz)');
```

```
xlabel('Time (s)');
```

```
ylabel('Amplitude');
```



Experiment 6.4

% Aliasing Effect Demonstration

```
clc;
```

```
clear;
```

```
close all;
```

```
% Signal frequency
```

```
fm = 10; % Hz
```

```
% Sampling frequency (below Nyquist rate)
```

```
fs = 15; % Hz
```

```
% Continuous-time signal
```

```
t = 0:0.001:1;
```

```
x = sin(2*pi*fm*t);  
% Sampled signal  
n = 0:1/fs:1;  
xs = sin(2*pi*fm*n);  
% Plotting  
figure;  
subplot(2,1,1);  
plot(t,x,'LineWidth',2);  
grid on;  
title('Original Continuous-Time Signal');  
  
xlabel('Time (s)');  
ylabel('Amplitude');  
subplot(2,1,2);  
stem(n,xs,'filled');  
grid on;  
title('Aliased Sampled Signal');  
xlabel('Time (s)');
```

```
ylabel('Amplitude');
```

